

MR-RAG: Multimodal Relevance-Aware Retrieval-Augmented Generation for Medical Visual Question Answering

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Abstract

Large Vision Language Models (LVLMs) with retrieval-augmented generation (RAG) are emerging as a main paradigm for processing vision-language medical tasks due to their promising achievements. However, existing approaches exhibit two significant limitations in both retrieval and generation stage: First, during the retrieval stage, most methods typically rely on a single similarity signal to estimate document relevance, ignoring the rich information available in multimodal data, which may fail to accurately retrieve matching content. Second, in the generation stage, retrieved documents are integrated directly and uniformly into the input for LVLMs, without taking into account their varying relevance to the question, which may result in the dilution of crucial information and exacerbate the negative impact of irrelevant content. To address these limitations, we propose MR-RAG, a dual-stage RAG enhancement framework by considering multimodal relevance in both retrieval and generation phases. Specifically, a Multimodal Cooperative Retrieval (MCR) module leverages intra- and cross-modal signals for precise document alignment. Subsequently, an Importance-Aware Information Flow Augmentation (IFA) mechanism adjusts attention paths based on fused relevance to refine answer generation. By bridging retrieval and generation via multimodal signals, MR-RAG significantly improves factual accuracy. Experiments on three medical datasets show our method outperforms state-of-the-art baselines, achieving up to a 6.4% accuracy gain.

1. Introduction

Recent advances in Large Vision Language Models (LVLMs) [5, 22, 38] have shown remarkable performance in a wide range of vision-language tasks, including visual question answering (VQA) [3, 10], visual reasoning [15],

and image captioning [16]. These models are rapidly becoming foundational tools for multimodal reasoning and are showing promising potential in specialized clinical or healthcare domain applications. [1, 20, 26, 36].

However, applying general-purpose LVLMs to the medical domain remains challenging due to a significant domain gap between their pretraining data and the specialized medical knowledge required for accurate and reliable predictions. To bridge this gap, retrieval-augmented generation (RAG) [19] has emerged as a promising technique. By retrieving relevant external knowledge and incorporating it into the model’s input, RAG empowers LVLMs to produce more accurate and informed medical responses [31, 32]. Recent studies have demonstrated the effectiveness of RAG in tasks such as medical visual question answering (VQA) [34, 36] and radiology report generation [4, 17, 28, 29], highlighting its potential to enhance factuality and domain alignment. While effective in principle, existing RAG approaches for medical vision-language tasks still suffer from two critical limitations: (1) **Single-similarity retrieval strategies**, which rely solely on one type of similarity signal and thus fail to capture the rich intra-modal and cross-modal semantics inherent in clinical data such as radiographs and medical reports; and (2) **Relevance-agnostic generation**, where retrieved documents are incorporated into the input without differentiation, making it difficult for the model to distinguish useful content from irrelevant or noisy evidence. These issues not only limit the precision of retrieved content but also weaken the effectiveness of document integration, ultimately leading to factually incorrect or misleading outputs that are especially risky in high-stakes medical decision-making.

To address these limitations in a unified manner, we propose **MR-RAG**, a dual-stage enhancement framework that leverages multimodal relevance signals throughout both retrieval and generation stages. In the first stage, we introduce a **Multimodal Cooperative Retrieval (MCR)** mechanism that leverages a lightweight adapter to jointly model

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intra-modal and cross-modal similarities, enabling more accurate estimation of multimodal relevance scores compared to those single-similarity retrieve manners. In the second stage, we propose an **Importance-Aware Information Flow Augmentation (IFA)** strategy that modulates the internal attention dynamics of the LVLM, amplifying information flow from more relevant documents while suppressing the influence of less useful ones during answer generation. By leveraging multimodal relevance at both stages, MR-RAG ensures a coherent and controllable selection and utilization of information throughout retrieval and generation, leading to more accurate and reliable responses in medical VQA task.

In summary, our main contributions are as follows:

- We propose **MR-RAG**, a unified dual-stage enhancement framework for retrieval-augmented generation in medical vision-language tasks, which leverages multimodal relevance signals across retrieval and generation stages.
- We propose a Multimodal Cooperative Retrieval (MCR) module that incorporates a lightweight adapter to leverage complementary intra-modal and cross-modal signals, which collectively enable more precise and comprehensive retrieval.
- We introduce an Importance-Aware Information Flow Augmentation (IFA) module that integrates document-level relevance into the attention mechanism of the LVLM, effectively prioritizing critical evidence while suppressing irrelevant content during answer generation.
- We conduct extensive experiments on three widely used medical VQA datasets: IU-Xray [7], Harvard-FairVLMed [25], and MIMIC-CXR [14], demonstrating that MR-RAG consistently outperforms strong baselines by a significant margin.

2. Related Work

2.1. Retrieval-Augmented Generation in Medical Large Vision Language Models

Retrieval-Augmented Generation (RAG) has emerged as an effective strategy to enhance the factuality and generalization of language models by incorporating external knowledge at inference time. In the medical domain, this paradigm has been adapted to Med-LVLMs [20, 26] to support complex tasks such as visual question answering and radiology report generation. Recent efforts have primarily focused on improving the retrieval stage of medical RAG systems. RAMM [34] enhances medical VQA by retrieving caption-style knowledge from PubMed Central. eCLIP [17] integrates expert annotations to mitigate the modality gap, thereby improving retrieval quality. MCSAM [29] introduces a memory-augmented retriever to support more coherent and informative medical summarization. More advanced systems like RULE [31] and MMed-RAG [32] aim

to jointly optimize both retrieval and generation stages: RULE proposes a calibrated retrieval strategy to balance content quantity and leverages preference fine-tuning to reduce over-reliance on retrieved medical reports. MMed-RAG builds a domain-aware retriever with adaptive context selection, coupled with RAG-specific preference alignment.

While effective, these methods do not explicitly model the importance of each retrieved document or inject this information into the model’s internal reasoning process. In contrast, our work introduces a fine-grained mechanism that estimates document-level importance based on intra-modal and cross-modal similarities, and injects these signals into attention layers to dynamically guide information flow, leading to more faithful and robust knowledge usage.

2.2. Attention Steering for Controllable Generation

Attention steering refers to techniques that explicitly manipulate the internal attention mechanisms of large language models (LLMs) during inference by selectively reweighting attention scores on critical tokens, thereby improving controllability and factual consistency without requiring model retraining. For example, ActAdd [30] steers model behavior by adding carefully crafted activation vectors. ITI [21] introduces steering vectors computed via trained probes and applies them selectively across attention heads to promote truthful outputs. PASTA [35] performs post-hoc attention reweighting to enhance the model’s focus on user-specified tokens. PAI [23] adaptively adjusts attention weights on image tokens in multimodal models to improve alignment between visual and textual modalities.

While promising, most of these methods are designed for either unimodal language generation or require heavy supervision. In contrast, RAG offers a built-in yet under-utilized form of supervision: the semantic relevance of retrieved documents. Our work bridges attention steering and retrieval augmentation by injecting relevance-based importance signals into attention layers, enabling adaptive and importance-aware information flow during generation.

3. Preliminaries

We begin by reviewing the architecture of LVLMs and their integration with RAG. We then introduce the attention mechanism, with a focus on the modeling of information flow within transformer layers.

3.1. Problem Formulation: LVLM-based RAG for Medical VQA

Let $q = (q_i, q_t)$ denote a multimodal medical question, where q_i is a radiographic or fundus image and q_t is the associated question. Let $\mathcal{D} = \{d^k = (d_i^k, d_t^k)\}_{k=1}^N$ denote a retrieval corpus of multimodal documents, each consisting of an image d_i and its corresponding medical report d_t . LVLM-based RAG follows a two-stage pipeline:

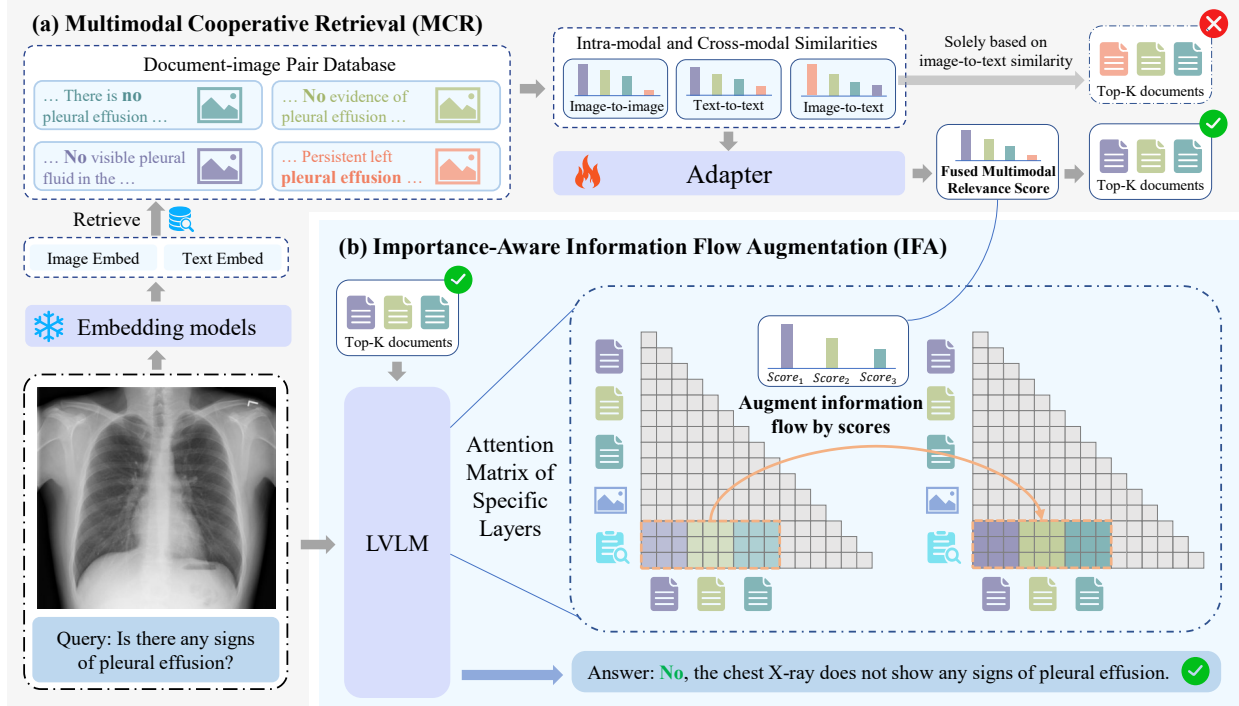


Figure 1. Overview of the proposed MR-RAG, consisting of: (a) **Multimodal Cooperative Retrieval (MCR)**, which computes a fused relevance score by combining cross-modal and intra-modal similarities for more accurate retrieval; (b) **Importance-Aware Information Flow Augmentation (IFA)**, which selectively enhances attention flow based on fused multimodal relevance to improve answer generation in medical VQA.

- **Retrieval Stage:** Given the query q and the retriever $\mathcal{R}$, retrieve a set of top- K documents $\mathcal{R}(q) = \{d^k\}_{k=1}^K \subset \mathcal{D}$ that are most relevant to the query.
- **Generation Stage:** The LVLM takes the query and the retrieved documents as input and generates an answer $\hat{a}$. Formally, the overall process is:

$$\hat{a} = \text{LVLM}(q, \mathcal{R}(q))$$

In this paradigm, both the **quality of the retrieved documents** and the model’s **ability to effectively utilize their information** are critical to overall performance.

3.2. Information Flow in Attention

Transformer-based LLMs rely heavily on the self-attention mechanism to capture dependencies among tokens. In each layer, the attention scores and outputs are computed as:

$$\tilde{A} = \frac{QK^\top}{\sqrt{d}}, \quad A = \text{softmax}(\tilde{A}), \quad O = AV$$

where $Q, K, V \in \mathbb{R}^{n \times d}$ are the query, key, and value matrices, and n is the number of input tokens. The attention map $A \in \mathbb{R}^{n \times n}$ contains the attention weights of all token pairs. Each entry A_{ij} represents the degree to which token

i attends to token j , which can be interpreted as a unit of information flow from token j to token i .

In RAG settings, input sequences include both query tokens and tokens from retrieved documents. However, standard attention mechanisms lack explicit supervision to distinguish between informative and noisy documents, which limits their ability to effectively model information flow.

4. Methodology

Our proposed approach MR-RAG, as shown in Figure 1, consists of two key components designed to address the limitations of conventional retrieval-augmented generation in the medical domain.

First, in the retrieval stage, to mitigate the issue of inaccurate or noisy retrieval results, we introduce a Multimodal Cooperative Retrieval (MCR) module. This module fully leverages the complementary strengths of different modalities to more precisely match relevant documents from the multimodal knowledge base.

Second, in the generation stage, we propose an Importance-Aware Information Flow Augmentation (IFA) mechanism. This component dynamically adjusts the contribution of each retrieved document by augmenting the attention flow based on its semantic relevance to the query.

As a result, the model is able to amplify the influence of critical evidence while suppressing irrelevant or misleading content, leading to more accurate and robust generation.

4.1. Multimodal Cooperative Retrieval

In existing medical RAG frameworks [12, 31, 32], the retrieval stage typically relies only on the similarity between the medical image in the query and medical reports in the knowledge base. However, this single-similarity approach overlooks other potentially informative modality pairs, such as image-to-image and text-to-text similarities, consequently restricting the system’s ability to exploit the full richness of multimodal information.

To address this limitation, we propose a **Multimodal Cooperative Retrieval (MCR)** (Figure 1 (a)), which jointly considers cross-modal and intra-modal similarity signals to improve retrieval quality. Given a query $q = (q_i, q_t)$ and a candidate document $d = (d_i, d_t)$ in the knowledge base, the fused relevance score is computed as:

$$f(q, d) = \alpha \cdot \text{Sim}(q_i, d_t) + \beta \cdot \text{Sim}(q_i, d_i) + \gamma \cdot \text{Sim}(q_t, d_t) \quad (1)$$

where $\text{Sim}(\cdot)$ denotes cosine similarity. The weights α , β , and γ are learnable parameters that adaptively reflect the relative importance of each modality. We optimize them using the InfoNCE loss [27]:

$$\mathcal{L} = -\log \frac{e^{s^+/\tau}}{e^{s^+/\tau} + \sum_j e^{s_j^-/\tau}}, \quad (2)$$

where s^+ is the fused relevance score for the positive candidate and s_j^- denotes those for negatives. InfoNCE loss encourages high similarity for positive pairs and low similarity for negatives, aligning with MCR’s objective of learning discriminative relevance scores. Training is performed on a lightweight contrastive dataset $\mathcal{D}_{\text{train}}$ constructed from retrieval candidates. See Algorithm 1 for the details on the training procedure.

This relevance formulation allows the retrieval module to integrate complementary cues from multiple modalities, enhancing both retrieval precision and robustness, particularly in clinical scenarios where relying solely on single-similarity may be insufficient.

4.2. Importance-Aware Information Flow Augmentation

In standard RAG pipelines, although the retrieval module provides the language model with a set of potentially relevant documents, these documents are often treated equally or simply ranked [9, 24, 33] without considering their semantic relevance during fusion. This uniform treatment fails to capture substantial differences in informativeness

Algorithm 1 Build Training Dataset and Train Adapter

Input: Retrieved candidate set $\mathcal{D}_{\text{can}}$, multimodal model $\mathcal{M}$, adapter weights (α, β, γ)

Output: Trained adapter weights (α, β, γ)

- 1: Initialize training set $\mathcal{D}_{\text{train}} \leftarrow \emptyset$
 - 2: **for all** $(q, a, \{d^k\})$ in $\mathcal{D}_{\text{can}}$ **do**
 - 3: Use $\mathcal{M}$ to get the correct confidence conf^k of all d^k
 - 4: Select top-1 positive as p and bottom- m negatives as $\{n^m\}$
 - 5: Add $(q, p, \{n^m\})$ to $\mathcal{D}_{\text{train}}$
 - 6: **end for**
 - 7: **for all** $(q, p, \{n^m\})$ in $\mathcal{D}_{\text{train}}$ **do**
 - 8: Compute positive score $s^+ = f(q, p)$ and negative scores $s_m^- = f(q, n^m)$ via Eq. 1
 - 9: Compute contrastive loss $\mathcal{L}$ using Eq. 2
 - 10: Update weights $(\alpha, \beta, \gamma) \leftarrow (\alpha, \beta, \gamma) - \eta \cdot \nabla \mathcal{L}$
 - 11: **end for**
 - 12: **return** (α, β, γ)
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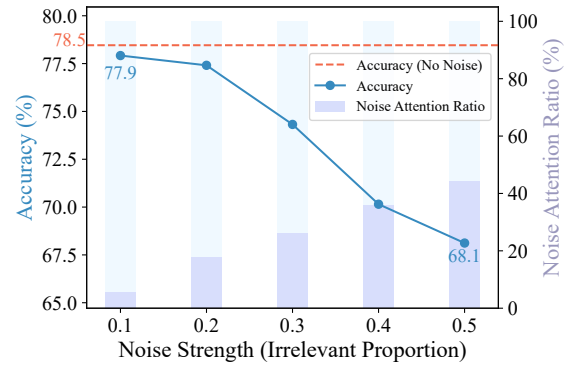


Figure 2. Impact of irrelevant document proportion on model reliability. Increasing irrelevant content raises noise attention ratio and reduces accuracy, revealing the model’s sensitivity to noisy evidence.

among the documents. This one-size-fits-all approach leads to two key issues:

- High-quality documents may contribute insufficiently due to diluted attention.
- Irrelevant or noisy documents may disproportionately influence the generation, introducing factual errors or semantic drift.

To investigate how irrelevant documents interfere with high-quality evidence, we conduct an experiment by mixing varying proportions of irrelevant documents into the retrieved set and inputting them to the model. We then observe the changes in answer accuracy and the distribution of attention over relevant and irrelevant content. As illustrated in Figure 2, with the increasing proportion of irrelevant content, the noise attention ratio rises while the accuracy drops

from 77.9% at a low noise level to 68.1% when irrelevant documents account for half of the input. This indicates that standard RAG models struggle to distinguish between relevant and irrelevant information, making them vulnerable to distraction by unreliable content.

This observation motivates the need for a mechanism that can dynamically suppress the effect of irrelevant documents and enhance the impact of truly relevant ones. Inspired by this, we consider integrating document relevance signals into the attention flow. However, arbitrarily adjusting attention pathways may lead to unintended effects. To better understand which components of the attention mechanism are crucial for accurate answer generation, we conduct attention blocking experiments by selectively masking interactions between different token types across transformer layers. Our findings reveal that the attention flow from retrieved document tokens to question tokens, which we refer to as **Reports**→**Question**, in certain key layers $\mathcal{L}$ plays a critical role in mediating high-quality information transfer, as detailed in the section 7.1.

Based on these findings, we propose the Importance-Aware Information Flow Augmentation (IFA) mechanism (Figure 1 (b)), which utilizes the relevance scores computed by the Multimodal Cooperative Retrieval (MCR) as document-level importance signals to dynamically steer information flow from retrieved documents to the query.

Specifically, for an attention logits matrix $\tilde{A}$ in the key layers $\mathcal{L}$, we modulate the cross-attention between document tokens and question tokens by adding a relevance-based bias λ_{ij} to produce the modulated logits $\tilde{A}'$:

$$\tilde{A}'_{ij} = \tilde{A}_{ij} + \lambda_{ij} \cdot |\tilde{A}_{ij}| \quad (3)$$

The modulation coefficient λ_{ij} is defined as:

$$\lambda_{ij} = \begin{cases} \frac{\text{Score}_k - \text{Score}_{\min}}{\text{Score}_{\max} - \text{Score}_{\min}}, & \text{if } i \in \mathcal{Q}, j \in \mathcal{D}_k \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

where $\mathcal{Q}$ and $\mathcal{D}_k$ denote the token sets of the query and the k -th retrieved document, respectively. The Score_k is the fused relevance score of k -th retrieved document. The Min-Max normalization operation maps scores to $[0,1]$, ensuring that λ_{ij} serves as a positive, relative weighting factor.

Finally, the modulated attention map is computed as:

$$A = \text{softmax}(\tilde{A}'), \quad (5)$$

which ensures that relevant documents to exert greater influence while irrelevant ones are relatively suppressed. By softly enhancing information flow along high-relevance paths, IFA improves semantic grounding and robustness in answer generation. Compared with hard masking, our soft modulation preserves model flexibility and avoids abrupt attention shifts. The full pipeline of our MR-RAG is shown in Algorithm 2.

Algorithm 2 Multimodal Cooperative Retrieval and Information Flow Augmentation (MR-RAG)

Input: Query $q = (q_t, q_i)$, retrieval set $\mathcal{D}_{\text{ret}}$, adapter weights (α, β, γ) , enhanced layers $\mathcal{L}$, model $\mathcal{M}$

Output: Predicted answer $\hat{a}$

- 1: Compute relevance scores $s = f(q, d)$ for all $d \in \mathcal{D}_{\text{ret}}$ using Eq. 1 with (α, β, γ)
 - 2: Select top-K $\{d_t^k\} \in \mathcal{D}_{\text{ret}}$ according to scores
 - 3: Concatenate (q_t, q_i) with $\{d_t^k\}$ to form input sequence $\mathcal{X}$ and feed into $\mathcal{M}$
 - 4: **for** each layer $l \in \mathcal{L}$ **do**
 - 5: Modify attention using Information Flow Augmentation (Eq. 3) based on $\{s^k\}$
 - 6: **end for**
 - 7: **return** Final output sequence $\hat{a}$
-

5. Experiment

In this section, we evaluate the effectiveness of our MR-RAG framework in multimodal medical question answering tasks. We first describe the experimental setups, including model configurations, baseline and evaluation datasets. Then, we present quantitative results on three real-world clinical datasets using standard evaluation metrics.

5.1. Experimental Setups

Implementation Details. We adopt LLaVA-Med-1.5 [20] as the backbone LVLm, which integrates a ViT-based vision encoder and a language decoder pretrained on medical instruction-following tasks. For document retriever, we employ a dual-encoder architecture: a ResNet-50 [11] vision encoder and BioClinicalBERT [2] as the text encoder. Given a multimodal query, we select the Top-15 candidates for downstream reasoning.

Baselines. We compare our proposed MR-RAG framework with three categories of strong baselines: **(1) Decoding-based hallucination mitigation methods:** including Beam Search [8], DoLa [6], OPERA [13], and VCD [18]. These methods focus solely on enhancing the decoding stage, without improving the retrieval process or modulating attention based on document relevance. **(2) Multimodal RAG-based methods:** including FactMM-RAG [28] and MMed-RAG [32], all of which adopt the same Med-LVLm backbone (LLaVA-Med-1.5) as ours but differ in retrieval strategies. **(3) Open-source Med-LVLms:** including Med-Flamingo [26] and MiniGPT-Med [1], which are pre-trained or fine-tuned on medical data but do not incorporate retrieval-augmented mechanisms.

Evaluation Datasets. We evaluate our method on three real-world medical VQA datasets: Harvard-FairVLMed [25], IU-Xray [7], and MIMIC-CXR [14]. Following RULE [31], the test sets consist of 4285, 2573

Table 1. Performance comparison of different methods on three medical VQA datasets. **Bold** denotes the best result; Underline denotes the second-best. Our method achieves the best results across all metrics.

Method	Harvard-FairVLMed(%)				IU-Xray(%)				MIMIC-CXR(%)			
	Acc	Prec	Recall	F1	Acc	Prec	Recall	F1	Acc	Prec	Recall	F1
LLaVA-Med-1.5	77.35	59.50	62.56	60.34	85.64	81.56	84.12	82.62	71.92	71.02	72.01	71.16
Decoding-based hallucination mitigation methods												
+Greedy	77.46	59.90	61.92	<u>60.42</u>	85.72	81.55	83.79	82.51	<u>72.93</u>	71.94	72.90	<u>72.13</u>
+Beam Search	71.51	58.19	<u>64.15</u>	58.17	84.76	80.56	81.36	80.93	71.75	72.07	<u>73.29</u>	71.45
+DoLa	75.88	57.93	59.50	58.13	85.92	81.88	85.19	83.17	70.91	70.63	71.78	70.42
+OPERA	73.94	57.37	61.01	57.80	86.49	82.31	<u>85.61</u>	83.64	70.39	71.37	72.44	70.21
+VCD	70.53	56.45	60.73	56.13	85.41	81.12	85.74	82.73	65.35	70.98	66.90	67.88
Multimodal RAG-based methods												
+FactMM-RAG	<u>79.59</u>	59.79	61.02	60.29	<u>86.85</u>	<u>82.79</u>	84.84	<u>83.70</u>	72.32	<u>72.47</u>	67.20	67.76
+MMed-RAG	70.57	56.73	61.12	56.39	84.57	80.16	84.20	81.63	70.36	69.94	70.93	69.81
MR-RAG (Ours)	84.47	67.61	66.05	66.78	88.13	84.50	86.01	85.19	79.35	79.80	77.56	78.61

and 3460 samples, respectively. IU-Xray and MIMIC-CXR consist of chest radiographs paired with diagnostic reports, while Harvard-FairVLMed contains fundus photographs with associated textual annotations. All datasets involve multimodal inputs and require joint reasoning over visual and textual information.

5.2. Main Results

Table 1 shows the overall performance of different models across three datasets. Our method consistently outperforms all baselines on all four evaluation metrics. Notably, it achieves an accuracy improvement of +4.9% on Harvard-FairVLMed, +1.3% on IU-Xray, and +6.4% on MIMIC-CXR compared to the strongest baseline. These results confirm the effectiveness of our proposed retrieval and augmentation strategy in enhancing factual consistency and domain alignment in medical VQA.

Table 2 presents the performance of our MR-RAG against open-source Med-LVLMs (Med-Flamingo [26] and MiniGPT-Med [1]) in accuracy and F1 score. Our method consistently outperforms all baselines, highlighting the effectiveness of retrieval augmentation and relevance-aware modulation. Complete results are provided in Appendix E.

To provide a more comprehensive evaluation of our method’s generation ability, we additionally perform a report generation task comparing various RAG-based approaches. The full results are presented in Appendix A.

6. Ablation

6.1. Effect of MCR and IFA Modules

To assess the individual contributions of the Multimodal Cooperative Retrieval (MCR) and Importance-aware Infor-

Table 2. Performance comparison with open-source Med-LVLMs on three medical datasets. Dataset names are abbreviated for brevity: FairVLMed = Harvard-FairVLMed, IU = IU-Xray, MIMIC = MIMIC-CXR.

Model	FairVLMed		IU		MIMIC	
	Acc	F1	Acc	F1	Acc	F1
Med-Flamingo	<u>78.34</u>	<u>65.38</u>	63.98	57.99	<u>78.02</u>	<u>74.82</u>
MiniGPT-Med	65.80	58.16	<u>72.57</u>	<u>62.98</u>	72.21	73.79
MR-RAG	84.47	66.78	88.13	85.19	79.35	78.61

mation Flow Augmentation (IFA) modules, we conduct ablation experiments on the performance across Harvard-FairVLMed, IU-Xray and MIMIC-CXR datasets. The results are presented in Table 3, and the evaluated model variants along with key observations are summarized below:

- **LLaVA-Med-1.5 (Baseline)** directly generates answers based on the concatenated question, image, and retrieved reports, without retrieval enhancement or attention modulation. It achieves an F1 score of **69.61**.
- **+MCR** replaces the retriever with our MCR module, integrating intra-modal and cross-modal similarities for multimodal retrieval. This improves F1 to **73.34**, highlighting the benefit of richer cross-modal retrieval.
- **+MCR+IFA (Full MR-RAG)** further applies IFA to reweight attention across retrieved documents during decoding. This leads to the highest F1 score of **75.33**, demonstrating that importance-aware attention refinement further enhances generation quality.

Table 3. Ablation study results averaged on across Harvard-FairVLMed, IU-Xray and MIMIC-CXR datasets.

Model Variant	Acc	Prec	Recall	F1
LLaVA-Med-1.5	77.57	68.83	71.56	69.61
+MCR	81.95	73.78	73.72	73.34
+MCR+IFA	83.66	75.90	74.88	75.33

6.2. Effect of Retrieval Strategy

To more comprehensively validate the advantage of multi-score fusion in our Multimodal Cooperative Retrieval (MCR) module, we conducted an additional ablation study comparing the fused multi-score similarity with each individual single-score similarity. As shown in Table 4, the adaptively fused multi-score strategy in MCR consistently surpasses all single-similarity baselines in both accuracy and F1 score. These results demonstrate that jointly modeling heterogeneous retrieval signals enables MCR to more effectively capture complementary cross-modal and intra-modal relevance cues.

Furthermore, among all single-score similarities, the $T2I$ ($\text{Sim}(q_t, d_i)$) score performs the weakest. This can be attributed to the weak semantic alignment in the medical VQA setting: textual questions (q_t) are often concise and abstract, whereas document images (d_i) contain dense and heterogeneous visual details. Such direct alignment between text and image introduces noise and instability. Given this semantic mismatch and its limited empirical contribution, we omit the $T2I$ term from the final MCR fusion to preserve model simplicity and enhance retrieval robustness.

Table 4. Comparison between single-score and multi-score retrieval strategies. MCR denotes our fused multi-score similarity.

Score	Accuracy	F1
I2I ($\text{Sim}(q_i, d_i)$)	75.09	66.13
I2T ($\text{Sim}(q_i, d_t)$)	<u>77.51</u>	<u>69.61</u>
T2T ($\text{Sim}(q_t, d_t)$)	73.05	65.51
T2I ($\text{Sim}(q_t, d_i)$)	64.71	60.88
MCR (Multi-Score Fusion)	81.95	73.34

6.3. Impact of Retrieval Size

We evaluate the performance of MR-RAG and baselines with other decoding methods across different retrieval sizes $K \in \{5, 10, 15, 20, 25, 30\}$. As shown in Figure 3, our method consistently outperforms all baselines by a significant margin under all settings.

More importantly, we observe that the performance of existing baselines fluctuates substantially as K varies. For example, the accuracy of beam search varies from 65.2%

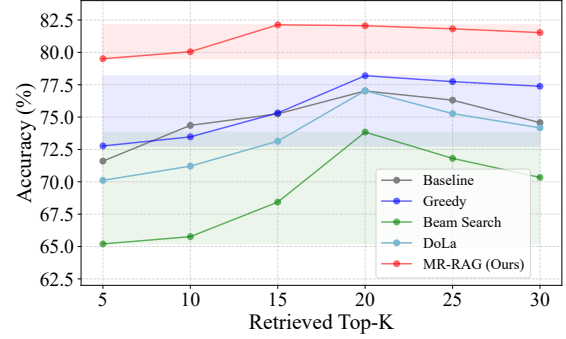


Figure 3. Accuracy of different methods under varying Top- K values on the Harvard-FairVLMed dataset. MR-RAG achieves both higher and more stable performance.

to 73.8% (a range of 8.6%), and the greedy decoding baseline varies by 4.5%. In contrast, our method only fluctuates within a narrow band of 2.6% ([79.5%, 82.1%]), demonstrating a clear robustness to the choice of retrieval size.

We attribute this stability to two key design features: (1) our MCR module more accurately identifies semantically relevant candidates, and (2) our IFA mechanism filters out unhelpful retrievals during answer generation. Together, they enable MR-RAG to consistently leverage precise evidence while mitigating the impact of noise, resulting in robust performance across varying retrieval sizes.

7. Analysis

7.1. Information Flow Blocking Analysis

To modulate attention flow effectively, we first pinpoint the critical layers and token-level interactions that drive factual reasoning. We perform a systematic layer-wise attention blocking analysis by selectively disabling attention weights along specific paths across transformer layers, revealing how information flows between modalities and sources affect the final prediction.

We categorize input tokens into four groups based on their source and position in the input sequence: **Reports tokens** (the text part of retrieved documents), **Image tokens** (the query image), **Question tokens** (the query text), and the **Last token** (where the model begins generating the answer). These categories represent the complete information chain required for multimodal reasoning.

Inspired by [37], we use a sliding window of size 7 over the transformer layers, progressively block the attention flow from one token group to another within the windowed layers. Blocking is performed by zeroing out the corresponding entries in the attention maps, thereby disabling information flow between selected token groups. We then observe the resulting change in answer accuracy to assess the importance of each pathway.

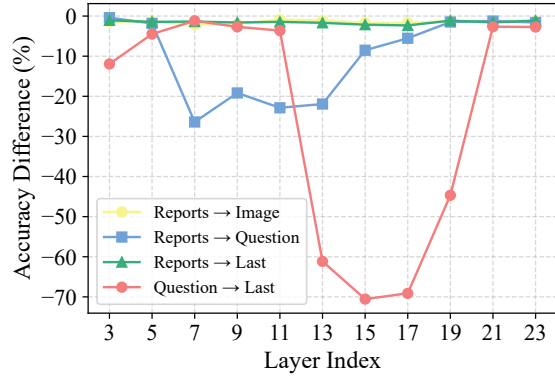


Figure 4. Accuracy drop caused by blocking attention between token groups at different layers, evaluated using LLaVA-Med-1.5 on the IU-Xray dataset. Blocking **Reports**→**Question** in shallow layers and **Question**→**Last** in deeper layers yields the most performance degradation.

As shown in Figure 4, two distinct stages of information flow emerge from our analysis:

- **Evidence gathering stage:** Blocking the information flow **Reports**→**Question** in shallow-to-mid layers leads to a significant accuracy drop. This suggests that the model relies on integrating content from the retrieved reports into the question tokens during this stage.
- **Answer consolidation stage:** Blocking the information flow **Question**→**Last** in mid-to-deep layers causes even more severe degradation, indicating that enriched question representations are consolidated into the answer token in the final layers.

In contrast, blocking other information flows such as **Reports**→**Image** or **Reports**→**Last** yields negligible change, implying that reports contribute little information via these paths and thus require no further enhancement.

Guided by this empirical insight, our Information Flow Augmentation module targets the **Reports**→**Question** path within layers 8–15, dynamically modulating attention strength based on the fused relevance of each retrieved report. Although the **Question**→**Last** path is also sensitive to blocking, it reflects a downstream flow from question to answer tokens that only indirectly incorporates retrieved data. Given that **Reports**→**Question** represents the primary conduit for direct evidence integration, we concentrate our enhancement efforts on this pathway to ensure achieving more grounded and precise generation.

7.2. Case Study

We present two representative examples to further illustrate how the proposed modules enhance both retrieval precision and information utilization within the RAG pipeline. As shown in Figure 5(a), we compare the retrieved documents with and without the Multimodal Cooperative Re-

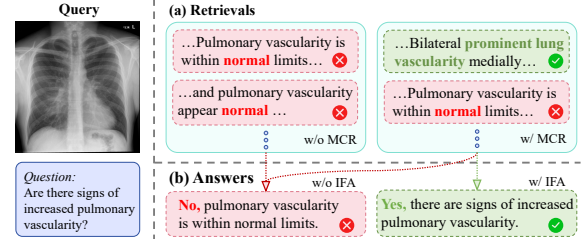


Figure 5. Case study illustrating the effect of MCR (a) and IFA (b). MCR improves retrieval precision by promoting the most relevant document. IFA enables the model to focus on relevant content and suppress noise during generation.

trieval (MCR) module. Without MCR, the most relevant evidence is missing from the top results. In contrast, MCR successfully elevates the correct document to the top position by leveraging intra-modal and cross-modal similarity, providing a more accurate and semantically grounded context for subsequent generation. Figure 5 (b) compares the generation outputs with and without the Importance-Aware Flow Augmentation (IFA) mechanism. While the mixed-quality documents retrieved by MCR can still lead the model to generate incorrect answers, the IFA module enables the model to attend more strongly to highly relevant documents while suppressing noisy ones, ultimately leading to a correct and faithful response.

8. Conclusion

In this work, we address two critical limitations in existing medical RAG systems: single-similarity retrieval and relevance-agnostic generation. By proposing MR-RAG. Our approach introduces a Multimodal Cooperative Retrieval (MCR) module and an Importance-Aware Information Flow Augmentation (IFA) mechanism, enabling the model to both *select* and *utilize* evidence in a relevance-aware manner. Extensive experiments on three medical VQA datasets demonstrate that our method significantly improves factual accuracy and robustness across varying settings. However, our current method focuses on modulating information flow between retrieved texts and the question tokens, while leaving the role of images largely unexplored. In future work, we aim to extend our approach to model cross-modal attention pathways, further strengthening the adaptability and explainability of RAG systems in complex multimodal reasoning tasks. Moreover, the general principles behind MR-RAG, comprising cooperative retrieval fusion and importance-aware attention modulation, are not confined to the medical domain. We believe these ideas can be readily transferred to other knowledge-intensive settings, such as legal or scientific question answering, where reliable retrieval and grounded generation are equally crucial.

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